MailFlow

Technical Architecture Document (TAD)

1. Introduction

This document describes the technical architecture of MailFlow, an AI-powered outreach platform that enables users to upload leads, perform AI company research, generate personalised emails, preview/edit content, send campaigns through a background queue, and track outreach.

2. System Overview

Workflow: Upload Leads → Column Mapping → Duplicate Detection → Database → AI Company Research → Email Generation → Preview & Edit → BullMQ Queue → Email Provider → Analytics.

Layer	Technology
Frontend	React + Vite + Tailwind CSS
Backend	Node.js + Express
Database	Supabase PostgreSQL
Queue	BullMQ + Redis
AI	OpenAI
Email	Resend / SendGrid

3. High-Level Architecture

Frontend (React+Vite) communicates with an Express REST API. PostgreSQL stores application data, Redis powers BullMQ queues, OpenAI generates research and emails, and Resend/SendGrid delivers emails.

4. Technology Stack

Frontend: React, Vite, Tailwind CSS Backend: Node.js, Express Database: PostgreSQL (Supabase) Queue: BullMQ + Redis AI: OpenAI Hosting: Vercel + Railway

5. Frontend Architecture

Pages, Components, Layouts, Services, Hooks and Utilities organised using a modular folder structure.

6. Backend Architecture

Controllers, Routes, Middleware, Services, Workers, Config and Utility modules.

7. Database

Users, Leads, LeadCustomFields, Campaigns, Templates, EmailJobs, ResearchCache and ActivityLogs.

8. AI Lead Intelligence

Reads website content, extracts company information, generates summary, industry, services and personalised context for outreach.

9. Email Engine

Combines template, lead data and AI research to generate unique emails. Users can preview, edit and regenerate before sending.

10. Queue Processing

Campaign emails are queued using BullMQ. Workers send emails with retries, rate limiting and failure handling.

11. WhatsApp Module

Creates personalised wa.me links. User reviews and manually sends messages.

12. Security

JWT authentication, bcrypt password hashing, HTTPS, input validation, Helmet, CORS, SQL injection protection and environment variables.

13. Deployment

Frontend on Vercel, Backend & Worker on Railway, PostgreSQL on Supabase, Redis, OpenAI API and Resend integration.

14. Scalability

Horizontal worker scaling, Redis queues, PostgreSQL indexing, caching and stateless backend services.

15. Future Enhancements

LinkedIn import, Chrome extension, AI follow-ups, WhatsApp Business API, team collaboration and CRM integrations.